

A Three-Dimensional Counterexample to Swift's $P(\omega_1)$

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2026-06-29

Source and Target

The source paper is Andrew Swift, *Coarse embeddings into $c_0(\Gamma)$* , arXiv:1703.01891. In Section 3 Swift defines $P(\alpha)$: for every $n \in \mathbb{N}$ and every coloring

$$\sigma : [\alpha]^n \rightarrow \mathcal{C},$$

at least one of the following alternatives holds:

1. there is a sequence $(F_j)_{j=1}^\infty$ of pairwise disjoint members of $[\alpha]^n$ such that $\sigma(F_i) = \sigma(F_j)$ for all i, j ;
2. there is $F \in [\alpha]^{n-1}$ such that

$$\sigma(\{F \cup \{\gamma\} : \gamma \in \alpha \setminus F\})$$

is infinite.

Problem 2 in Section 5 asks whether Theorem B is true for ω_1 , equivalently whether $P(\omega_1)$ holds.

Combining (12) and (13) we finally obtain that for some $D \geq 1$, and any $k \in \mathbb{N}$,

$$\frac{1}{D} \max_j |a_j| \leq \left\| \sum_{i=1}^k a_i e_{\gamma_i} \right\| \leq D \max_j |a_j|. \quad (14)$$

for all $\{\gamma_1, \dots, \gamma_k\} \subset [1, \Lambda)$, which proves our claim. $\square$

5 Final comments and open problems

Let us mention in this final section some problems of interest.

First of all, we do not know whether or not Theorem A is true if we replace λ by smaller cardinal numbers.

Problem 1. Assume that X is a Banach space with $\text{dens}(X) \geq \omega_1$, and assume that X coarsely embeds into $c_0(\Gamma)$ for some cardinal number Γ . Does X have trivial co-type? If moreover X has a symmetric basis, must it be isomorphic to $c_0(\omega_1)$?

Of course Problem 1 would have a positive answer if the following is true.

Problem 2. Is Theorem B true for ω_1 ?

Connected to Problems 1 and 2 is the following

Result

The answer to Problem 2 is negative. The counterexample already appears in arity $n = 3$.

Theorem 1. *There is a coloring $\sigma : [\omega_1]^3 \rightarrow \omega_1$ such that:*

1. *for every $F \in [\omega_1]^2$, the set*

$$\{\sigma(F \cup \{\gamma\}) : \gamma \in \omega_1 \setminus F\}$$

is finite;

2. *no two disjoint triples have the same color.*

Consequently $P(\omega_1)$, and hence Theorem B with $\lambda = \omega_1$, is false.

Proof Intuition

The point is to exploit the fact that every proper initial segment of ω_1 is countable. For each top vertex γ , enumerate the smaller ordinals below γ in type ω . Given a triple $\alpha < \beta < \gamma$, look only at the two smaller vertices and color the triple by the one that occurs earlier in the enumeration of γ .

This makes every color class a star: if a triple has color ξ , then ξ is one of its vertices. Hence two same-colored triples always meet. On the other hand, for a fixed pair $\{\xi, \eta\}$, the only possible colors outside the pair arise from ordinals which occur before ξ in the fixed enumeration of η , and there are only finitely many of those.

Construction and Proof

For each nonzero $\gamma < \omega_1$, choose a bijection

$$e_\gamma : \mathbb{N} \rightarrow \gamma.$$

Let $r_\gamma : \gamma \rightarrow \mathbb{N}$ be its inverse, so $r_\gamma(\xi)$ is the position of ξ in the enumeration e_γ . Define a coloring $\sigma : [\omega_1]^3 \rightarrow \omega_1$ as follows. If

$$\alpha < \beta < \gamma < \omega_1,$$

put

$$\sigma(\{\alpha, \beta, \gamma\}) = \begin{cases} \alpha, & r_\gamma(\alpha) < r_\gamma(\beta), \\ \beta, & r_\gamma(\beta) < r_\gamma(\alpha). \end{cases}$$

The two cases exhaust all possibilities because r_γ is injective.

Pair-local finiteness

Fix a pair $\{\xi, \eta\} \in [\omega_1]^2$ with $\xi < \eta$. Let

$$E_{\xi, \eta} = \{\delta < \eta : r_\eta(\delta) < r_\eta(\xi)\}.$$

This set is finite, since only finitely many natural numbers are smaller than $r_\eta(\xi)$.

We claim that every triple extending $\{\xi, \eta\}$ has color in the finite set

$$\{\xi, \eta\} \cup E_{\xi, \eta}.$$

Let $\delta \in \omega_1 \setminus \{\xi, \eta\}$.

If $\delta > \eta$, then the triple is $\xi < \eta < \delta$, and by definition its color is either ξ or η .

If $\xi < \delta < \eta$, then the triple is $\xi < \delta < \eta$. Its color is either ξ or δ . In the latter case $r_\eta(\delta) < r_\eta(\xi)$, so $\delta \in E_{\xi, \eta}$.

If $\delta < \xi$, then the triple is $\delta < \xi < \eta$. Its color is either δ or ξ . Again, if the color is δ , then $r_\eta(\delta) < r_\eta(\xi)$, so $\delta \in E_{\xi, \eta}$.

Thus the color set on all extensions of the fixed pair $\{\xi, \eta\}$ is finite. Therefore the second alternative in $P(\omega_1)$ fails for this coloring and $n = 3$.

No disjoint monochromatic triples

For every triple $T \in [\omega_1]^3$, the color $\sigma(T)$ is one of the elements of T . Therefore, if two triples T and T' have the same color ζ , then

$$\zeta \in T \cap T'.$$

In particular, no two same-colored triples are disjoint. Hence there cannot be an infinite sequence of pairwise disjoint triples of one color. The first alternative in $P(\omega_1)$ also fails.

Both alternatives fail for $n = 3$, so $P(\omega_1)$ is false. □

Verification Notes

The only choice used is a countable enumeration of each countable ordinal $\gamma < \omega_1$, which is available in ZFC. The proof does not use CH or any additional set-theoretic hypothesis.

This counterexample is compatible with the separate partial packet showing that the $n = 2$ case of $P(\omega_1)$ is true. Thus the first possible failure occurs at arity 3.

Novelty and Search Bounds

Before packaging, the run indexes were searched for arXiv id 1703.01891, Swift’s title, $P(\omega_1)$, Theorem B for ω_1 , and Delta-system terminology. The only run hit was the preceding $n = 2$ partial packet. Local corpus searches found Swift’s source paper and later references to related coarse-embedding and Q^{ω_1} material, but no local solution packet or later paper resolving Problem 2. A bounded web search for combinations of “Coarse embeddings into c_0 ”, “Problem 2”, “Theorem B”, “ $P(\omega_1)$ ”, and “counterexample” did not surface an exact answer.

Human-Review Recommendation

Review as a likely valid full negative answer to Problem 2. The key checks are: (i) the arity $n = 3$ really suffices to refute $P(\omega_1)$, (ii) the fixed-pair extension color set is finite in all three relative-order cases, and (iii) same-colored triples all contain their color ordinal.

References

- [1] A. Swift, *Coarse embeddings into $c_0(\Gamma)$* , arXiv:1703.01891.